MATH 114 - Computational Stochastics

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Contents

1 Overview

1.1 Introduction

Stochastic models describe real life phenomena like motion of the cell, shape of polymers stock prices, and more. What can we compute analytically and numerically from these models?

Key questions:

1. What is the distribution of the system?
2. How can we sample from the distribution?
3. How can we compute mean, variance, and higher moments of these systems?

Main topics:

1. Sampling RVs
2. Markov chain Monte Carlo (MCMC)
3. Stochastic differential equations (numerical & computation)

1.2 Sampling

- Generating "hard" RVs from "easy" RVs.
- Compute "hard" integrals.

$\boxed{ex}$ Area of the unit circle, $A = \pi$
Let $U_1 \sim U(-1, 1)$ and $U_2 \sim U(-1, 1)$
Let $x = \begin{cases} 0 \text{ if } (u_1, u_2) \text{ outside circle} \\ 1 \text{ if } (u_1, u_2) \text{ inside circle} \end{cases}$
$\implies P(X = 1) = \frac{A}{4}, P(X = 0) = 1 - \frac{A}{4}$
$\mathbb{E}(X) = \frac{A}{4}$
After N trials we generate $X_1, X_2, ..., X_n$ iid
By the strong law of large numbers

$$\lim_{N \to \inf} \frac{1}{N} \sum_{i=1}^{N} X_i = \frac{A}{4} = \frac{\pi}{4}$$

1.3 MCMC

Construct Markov chains to generate target distribution.
$\boxed{ex}$ Ising model

2 Markov Chains

2.1 Lecture 9

Def: A stochastic process is a family of RVs, $X_\epsilon \in S$, indexed by t, where $S \subseteq \mathbb{R}^d$ is a state space. Typically, $t \in [0, \infty)$, or $t\ in[a,b]$, or $t \in \{0, 1, 2, 3, ...\}$, etc.

- A discrete time stochastic process, is a sequence of RVs, $X_n \in S$, $n = 0, 1, 2, ...$ or $n = 1, 2, 3, ...$
- If S is finite or countably infinite then the stochastic process is a discrete state process.

First, we will examine discrete-time, discrete pace, stochastic process.
Two basic questions:

- Find the joint distribution

$$P(X_0 = i_0, X_1 = i_1, ..., X_n = i_n)$$
$$i_0, i_1, ..., i_n \in S$$

- Find the long-time behavior of the distribution of X_n:

$$P(x_n = i), i \in S, n >> 1$$

Def: Markov processes

- A stochastic process $\{X_n\}_{n=0}^{\infty}$ is a Markov process if

$$P(X_{n+1} = i_{n+1} | x_0 = i_0, ..., x_n = i_n)$$
$$= P(X_{n+1} i_{n+1} | X_n = i_n)$$
$$\forall n \geq 0, \forall i_0, ..., i_{n+1} \in S$$

- A Markov process is time homogeneous if:

$$\begin{aligned}&P(x_{n+1} = j | x_n = i)\\&= P(X_1 = j | X_0 = i)\\&\forall n \geq 0, \forall i, j \in S\end{aligned}$$

  Here we consider discrete time/state, time homogeneous Markov Chains (MC).

<u>Def:</u> Transition probability

$$\begin{aligned}P(i, j) &= P(X_1 = j | X_0 = i)\\&= P(i \to j) \quad \forall i, j \in S\end{aligned}$$

<u>Def:</u> Transition matrix

$P = [p(i, j)]$ if $S = \{1, 2, 3, ...\}$ or $\{0, 1, 2, ...\}$
Here i,j are states and also are row/column labels

For any $n \geq 1, i_0, ..., i_n \in S$, we have:

$$\begin{aligned}&P(X_0 = i_0, ..., X_n = i_n)\\&= P(X_0 = i_0) P(X_1 = i_1 | X_0 = i_0) P(X_2 = i_2 | X_1 = i_1, X_0 = i_0)\\&... \ P(X_n = i_n | X_{n-1} = i_{n-1}, ..., X_0 = i_0) \qquad \text{[conditional probability]}\end{aligned}$$

$\boxed{ex}$ Let $X_0, X_1, ..., X_n \in S$ iid. Then $\{X_n\}_{n=0}^{\infty}$ is a time homogeneous MC. Let $S = \{1, 2, ..., N\}$. Then $p(i, j) = \mathbb{P}(X_1 = j | X_0 = i) = \mathbb{P}(X_1 = j) = p_j$

$$\implies p = \begin{bmatrix} p_1 & \cdots & p_n \\ \vdots & \ddots & \vdots \\ p_1 & \cdots & p_n \end{bmatrix}, \sum_{i=1}^{n} p_i = 1$$

2.2 Lecture 10

Recap:

- Discrete time/state, time homogeneous
- MC: $\{X_n\}_{n=0}^{\infty} \in S$

- $\mathbb{P}(X_{n+1} = i_{n+1} | X_0 = i_0, ..., X_n = i_n) = \mathbb{P}(X_{n+1} = i_{n+1} | X_n = i_n)$
  $= p(i_n, i_{n+1})$

- Transition probability $\mathbb{P}(X_1 = j | X_0 = i)$
  $= p(i, j) = p_{ij} \quad \forall i, j \in S$

- Transition matrix $\mathbb{P}[p(i, j)]$

$\boxed{ex}$ Two-state MC, e.g. state of phone $X_n \in S = \{0, 1\}$

$$\begin{cases} X_n = 0 & \text{phone free at time n} \\ X_n = 1 & \text{phone busy at time n} \end{cases}$$

$\mathbb{P}(\text{free} \to \text{busy in time step}) = p$
$\mathbb{P}(\text{busy} \to \text{free in time step}) = q$
$\mathbb{P}(\text{free} \to \text{free in time step}) = 1 - p$
$\mathbb{P}(\text{busy} \to \text{busy in time step}) = 1 - q$

Transition matrix:

$$P = \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix}$$

e.g. $p = \frac{1}{2}, q = \frac{1}{3}$

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

Let $\mathbb{P}(X_0 = 0) = \frac{1}{5}, \mathbb{P}(X_0 = 1) = \frac{4}{5}$
Then $\mathbb{P}(X_0 = 0, X_1 = 1, X_2 = 0)$
$= \mathbb{P}(X_0 = 0) \cdot p(0, 1) \cdot p(1, 0)$
$= \frac{1}{5} \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{30}$
And $\mathbb{P}(X_0 = 0, X_2 = 0) = \sum_{i=0}^{1} \mathbb{P}(X_0 = 0, X_1 = i, X_2 = 0)$

$\boxed{ex}$ $P_0 = I$, $p_0(i, j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$
$P_1 = P$, $p_1(i, j) = p(i, j)$

The distribution vector π_n for X_n is

$$\pi_n = (\pi_n(0), \pi_n(1), ...)$$
$$\text{where } \pi_n(i) = \mathbb{P}(X_n = i), i \in S$$
$$\text{Note that } \pi_n \text{ is a probability vector}$$

Property: Let $X_n \in S = \{1, 2, 3, ...\}, n \geq 0$ be the time homogeneous Markov Chain with transition matrix P. Then

- P is a stochastic matrix
- $P \cdot \mathbb{I} = \mathbb{I}$ where $\mathbb{I} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$
- 1 is an eigenvalue of P with eigenvector $\mathbb{I}$
- $P_{m+n} = P_m \cdot P_n$ (Chapman-Kolmogorov equation)
- $P_n = P^n$
- $\pi_n = \pi_0 P_n = \pi_0 P^n$
- $\pi_{n+1} = \pi_n P, \ \forall n \geq 0$

Proof: direct verification
e.g. $\mathbb{P}(X_1 = j) = \sum_{i \in S} \mathbb{P}(X_0 = i)\mathbb{P}(X_1 = j | X_0 = i)$
$\pi_1(j) = \sum \pi_0(j) P_{ij}$

Long-run behavior, invariant distribution

Def: Let $P = [p(i, j)]$ be the transition matrix for Markov Chain $\{X_n\}_{n=0}^{\infty}$ with $S = \{1, 2, ..., N\}$. A probability vector $\pi \in \mathbb{R}^n$ is a stationary or invariant distribution if $\pi^p = \pi$

2.3 Lecture 11 - Invariant Distribution

- Recap: Discrete time/state, time homogeneous Markov Chains. Transition matrix $P = [p(i, j)]$, $p(i, j) = \mathbb{P}(X_1 = j | X_j = i)$, $p_n(i, j) = \mathbb{P}(X_n = j | X_0 = i)$
- Markov Chains $\rightarrow$ MCMC

$\boxed{ex}$ 3 state MC

$$P = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{8} & \frac{2}{3} & \frac{5}{24} \\ 0 & \frac{5}{6} & \frac{1}{6} \end{bmatrix}$$
$$S = \{1, 2, 3\}$$
$$\mathbb{P}(X_1 = 1 | X_0 = 1) = \frac{3}{4}, \text{etc.}$$
$$\lim_{n\to\infty} P^n = \begin{bmatrix} \pi \\ \pi \\ \pi \end{bmatrix}, \ \pi = (\frac{2}{11}, \frac{4}{11}, \frac{5}{11})$$
$$\text{Set } \pi_0 = (\mathbb{P}(X_0 = 1), \mathbb{P}(X_0 = 2), \mathbb{P}(X_0 = 3))$$
$$= \mathbb{P}(X_0 = i)$$

Then

$$\begin{aligned} \pi_n &= \pi_0 p^n \\ \lim_{n\to\infty} \pi_n &= \pi_0 \lim_{n\to\infty} p^n \\ &= \pi_0 \begin{bmatrix} \pi \\ \pi \\ \pi \end{bmatrix} \\ &= \left(\pi(1) \cdot \sum_i \pi_0(i),\ \pi(2) \cdot \sum_i \pi_0(i), ...\right) \\ &= \pi \end{aligned}$$

<u>Remarks</u>

- $\lim_{n\to\infty} \mathbb{P}(X_n = j) - \pi(j) \forall j$
-
$$\begin{aligned} \pi_n &= \pi_{n-1} P \\ \lim_{n\to\infty} \pi_n &= \lim_{n\to\infty} \pi_{n-1} P \\ \pi &= \pi P \end{aligned}$$
- π is a left eigenvector of P with eigenvalue 1.

Def: Let $P = [p(i,j)]$ be the transition matrix for MC $\{X_n\}_{n=0}^{\infty}$ with $S = \{1, ..., N\}$. A probability vector $\pi \in \mathbb{R}^n$ is a stationary or invariant distribution for X_n (or P) if: $\pi = \pi P$

Remarks
Above example shows $\lim_{n\to\infty} \pi_n =$ invariant distribution (with some conditions).

Questions: invariant distribution.

- When do they exist?
- Are they unique?
- Roughly: typically have unique invariant distribution unless reducible or periodic.

Def: A finite state time-homogeneous MC $X_n (n \geq 0)$ with transition matrix P is regular if there exists an integer $k \geq 1$, such that $P^k > 0$, (i.e. all entries of P^k are strictly > 0).

Thm
Let $X_n (n \geq 0)$ be a regular N-state Markov Chain with transition matrix P. Then:

1. $\{X_n\}_{n=0}^{\infty}$ has a unique stationary distribution π, $\pi > 0$
2. $\lim_{n\to\infty} P^n = \begin{bmatrix} \pi \\ \vdots \\ \pi \end{bmatrix}$, i.e. $\lim_{n\to\infty} p_n(i,j) = \pi(j)$
3. For any initial distribution π_0, $\lim_{n\to\infty} \pi_0 P^n = \pi$ i.e. $\lim_{n\to\infty} \mathbb{P}(X_n = j) = \pi(j)\ \forall j$

Remarks

- 2. $\implies \lim_{n\to\infty} \mathbb{P}(X_n = j | X_0 = i) = \pi(j)$ independent of i

- From 2.:

$$\begin{aligned}
\lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1}p_k(i,j) &= \lim_{n\to\infty}p_n(i,j)=\pi(j)\\
\text{And } \frac{1}{n}\sum_{k=0}^{n-1}p_k(i,j) &= \frac{1}{n}\sum_{k=0}^{n-1}\mathbb{P}(X_k=j|X_0=i)\\
&= \frac{1}{n}\sum_{k=0}^{n-1}\mathbb{E}[\mathbb{I}[X_k=j]|X_0=i]\\
&= \mathbb{E}\Big[\frac{1}{n}\sum_{k=0}^{n-1}\mathbb{I}(X_k=j)|X_0=i\Big]\\
&\to \pi(j) \text{ as } n\to\infty
\end{aligned}$$

Which means the mean fraction of time spent in $J \to \pi(j)$

How do we find π?

- Solve $\pi P = \pi$
- Diagonalize $P = U\Lambda U^{-1}$

$$\begin{aligned}
\Rightarrow \lim_{n\to\infty}P^n &= \lim_{n\to\infty}(U\Lambda U^{-1})^n\\
&= U(\lim_{n\to\infty}\Lambda^n)U^{-1}\\
&= \begin{bmatrix}\pi\\ \vdots\\ \pi\end{bmatrix}
\end{aligned}$$

2.4 Lecture 12 - MCMC

Recap:

- Discrete time/space MC $\{X_n\}_{n=0}^{\infty}, X_n \in S$
- $P=[p(i,j)], p(i,j)=\mathbb{P}(X_1=j|X_0=i)$
- $\pi_0=\mathbb{P}(X_0=i),\ \pi_n=\mathbb{P}(X_n=i)\quad$ (pmf X_n)

- Stationary distribution π s.t. $\pi P = \pi$
- Regular: $\exists k \geq 1$ s.t. $P^k > 0 \Rightarrow$ unique stationary distribution $\pi > 0$

  And $\lim_{n \to \infty} P^n = \begin{bmatrix} \pi \\ \vdots \\ \pi \end{bmatrix}$, $\lim_{n \to \infty} \pi_n = \pi$
- Detailed balance: probability vector λ s.t. $\lambda_i p(i,j) = \lambda_j p(j,i)\ \forall i,j \in S$ $\Rightarrow \lambda P = \lambda \Rightarrow \lambda$ stationary with respect to P.

Remarks

1. Can view MC as evolution of probability vectors (pmfs): $\pi_0, \pi_1, ...$ with $\pi_{n+1} = \pi_n P$. This is hard to compute for $|S|$ large.
2. MC is a random sequence of states $X_0, X_1, ... \in S$ which is often easier to compute.

Recall: Discrete Inversion (Lecture 6)
e.g.: Generate discrete RV X

$$\begin{aligned} &P(X = i) = p_i,\ i = 1,2,3 \\ &p_1 + p_2 + p_3 = 1 \end{aligned}$$

1. $U \sim U[0,1]$
2. $k = min\{m | \sum_{i=1}^{m} p_i \geq U\}$
3. Set $X \leftarrow k$

Algorithm: Generate / simulate MC
Input: transition matrix P, initial distribution π_0
Output: MC $X_0, X_1, ... \in S, X_n \sim \pi_k = \pi_0 P^k$

1. Generate $X_0 \sim \pi_0$
2. For $k = 1, ..., n$, generate X_{n+1} from $p(X_k)$. i.e. pmf over S, given by the X_k'th row of P.
3. Repeat n times to get samples from π_k

MCMC: (R&K ch.6)

- Key idea: generate distribution π by simulating MC with stationary distribution π
- Often π is known up to a normalizing constant.
- $\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \mathbb{I}\{X_k = i\} = \pi(i)\ \forall i \in S$
- $\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} g(X_k) = \mathbb{E}_\pi(g(x))$

We assume that

- RV $X \in S$ is finite, e.g. $S = \{1, 2, ..., N\}$
- Target pmf of X is $\pi = (\pi(1), ..., \pi_N), \pi(i) = \mathbb{P}(X = i)\ \forall i \in S$
- π has special form $\pi(i) = \frac{1}{z} b(i) > 0\ \forall i \in S$.
  Where $b(i) > 0$ are known and $z > 0$ is unknown norm.

$$\sum_{i\in S} \pi(i) = 1 \iff \sum_{i\in S} b(i) = z$$

Remarks

- z is hard to compute, (sum of $N = |S|$ terms)
- z is called a partition function

Algorithm: Metropolis

- First construct proposal transition matrix $Q = [q(i,j)]$, where Q is symmetric.
- Choose initial state $X_0 \in S$ (e.g. randomly).

Given $X_k = i \in S$

1. Sample $Y \in S$ from $q(i, \cdot)$, i.e. $\mathbb{P}(Y = y | X_k = i) = q(i, y)\ \forall y \in S$
   Assume $Y = j$ is observed.
2. Let $\alpha = \alpha_{ij} = \min\left(1, \frac{\pi(j)}{\pi(i)}\right) \in [0, 1]$ and generate $U \sim U[0, 1]$.
   If $U \leq \alpha$, accept Y, $X_{k+1} \leftarrow Y$
   If $U > \alpha$, reject Y, $X_{k+1} \leftarrow X_k$
   (Accept Y with probability α)
   Output: $X_0, X_1, X_2, ...$

Remarks

- $\{X_n\}_{n=0}^{\infty}$ is MC with $\lim_{n\to\infty} X_n \sim \pi$
- Can choose $Q = [q(i,j)]$ to be simple, e.g. uniform jumps, $q(i,j) = \frac{1}{N}$
- Only need the ratios $\frac{\pi(j)}{\pi(i)}$, so we don't need z.
- Call Y a proposed state.
- Call α the acceptance probability.
- In practice, $\alpha = a_{ij}$ is easy to compute.
- Can be extended to non-symmetric Q

$\boxed{ex}$ 1D Ising model

$$\begin{aligned} S &= \{\sigma = (\sigma_1, ..., \sigma_m) | \sigma_i = -1 \text{ or } +1,\ 1 \le i \le m\} \\ &= \{-1, +1\}^m \\ |S| &= 2^m \text{ states} \end{aligned}$$

2.5 Lecture 13 - Metropolis

Metropolis Algorithm

$$S = \{1, 2, ..., N\}$$

Goal: Sample random variable X from some target distribution $\Pi(i) = \mathbb{P}(X = i) \implies \Pi_i(i) = \frac{1}{z} b(i)$

We have $Q = [q(i,j)]$ which is our transition matrix (symmetric).

1. Choose $X_0 \in S$. Given $X_n \in S$, sample $Y \in S$ from $q(X_n, i)$, i.e. $\mathbb{P}(Y = y | X_n = i)$
2. Set $\alpha = \min(1, \frac{\pi(y)}{\pi(X_n)}) \in [0,1]$.
   Generate $U \sim U[0,1]$
   If $U \le \alpha$, accept $Y : X_{n+1} \leftarrow Y$
   Otherwise, $U > \alpha$, reject Y, $X_{n+1} \leftarrow X_n$

1D Ising Model
$S = \{\sigma = \{\sigma_1, \sigma_2, ..., \sigma_n\} : \sigma_j = \pm 1\}$
$\Rightarrow 2^m$ states.

$$\begin{aligned}
&\Pi(\sigma) = \frac{1}{z} e^{-\beta \mathcal{H}(\sigma)} \\
&\mathcal{H} = -J \sum_{j=1}^{m-1} \sigma_j \sigma_{j+1} - h \sum_{j=1}^{m} \sigma_j \\
&J, h > 0
\end{aligned}$$

Large $\Pi(\sigma) \longleftrightarrow$ Low energy states.
Calculate physical quantities.

$$\begin{aligned}
m(\sigma) = <\sigma> &= \frac{1}{m}(N_+ - N_-) \\
&= \frac{1}{m} \sum (N_+ - N_-) \Pi_p(\sigma)
\end{aligned}$$

$$\frac{\Pi_\beta(y)}{\Pi_\beta(X_n)} = e^{-\beta(\Delta \mathcal{H})}$$

a. $1 < i < m$
$\Delta \mathcal{H} = 2J\sigma_i(\sigma_{i+1} + \sigma_{i-1}) + 2h\sigma_i$

2.6 Lecture 14

Recap: Metropolis Chain
We have

- State space $S = \{1, 2, ..., N\}$
- Proposed transition matrix $Q = [q(i,j)]$ which is symmetric
- Target distribution $\pi > 0$

Metropolis summary

- Given $X_k \in S$, choose y sampled from $q(X_k, \cdot)$, i.e. X_k'th row of Q
- Accept Y with probability $\min(1, \frac{\pi(y)}{\pi(X_k)})$

- Output: MC $X_0, X_1, ..., \in S$ with $X_n \sim \pi$ as $n \to \infty$

Convergence

- How big should n be to get $X_n \sim \pi$ 'roughly'?
- Hard to compute!
- Heuristic: Consider $R(n) = \text{Cov}(X_n, X_0)$

Thm

1. The transition matrix $P = [p(i,j)]$ of the Metropolis chain is

$$p(i,j) = \begin{cases} q(i,j)\min(1, \frac{\pi(j)}{\pi(i)}) & i \neq j \\ 1 - \sum_{k \in S, k \neq i} q(i,k)\min(\frac{\pi(k)}{\pi(i)}) & i = j \end{cases} \forall i,j \in S$$

2. π and P satisfy detailed balance (DB)

$$\pi(i)p(i,j) = \pi(j)p(j,i) \ \forall i,j \in S$$

Hence π is stationary

3. If $q(i,j) > 0 \forall i,j \in S$ and $p(i,i) > 0 \ \forall i \in S$ Then π is a unique stationary distribution and $\lim_{n\to\infty} \pi_0 P^n = \pi$

Proof

1. For $i \neq j$

$$\begin{aligned} p(i,j) &= \mathbb{P}(\text{propose } j|i) \cdot \mathbb{P}(\text{accept } j) \\ & q(i,j)\min(1, \frac{\pi(j)}{\pi(i)}) \end{aligned}$$

Then

$$\begin{aligned} &\sum_{j \in S} p(i,j) = 1 \ \forall i \in S \\ &\Rightarrow p(i,j) = 1 - \sum_{k \in S, k \neq i} q(i,k)\min(1, \frac{\pi(k)}{\pi(i)}) \end{aligned}$$

2. Want to show: $\pi(i)p(i,j) = \pi(j)p(j,i)\ \forall i,j \in S$.
   This is always true for $i = j$
   For $i \neq j$:

$$\begin{aligned}
\pi(i)p(i,j) &= \pi(i)q(i,j)\min(1, \frac{\pi(j)}{\pi(i)}) \\
&= q(i,j)\min(\pi(i), \pi(j)) \\
&= q(j,i)\min(\pi(j), \pi(i)) \\
&= \pi(j)p(j,i) \Rightarrow DB\checkmark \\
&\pi \text{ is stationary (Lecture 11)}
\end{aligned}$$

3. Conditions $\Rightarrow p(i,j) > 0\ \forall i,j \in S \Rightarrow P$ is regular
   $\Rightarrow \pi$ is unique and $\lim_{n\to\infty} \pi_0 P^n = \pi$

Remarks

- For Metropolis Algorithm applied to Ising model, we had:

$$S = \{\underline{\sigma} = (\sigma_1, ..., \sigma_n), \sigma_j = \pm 1\}$$

  Spin flips $(\sigma_1, ..., \sigma_j, ..., \sigma_n) \mapsto (\sigma_1, ..., \sigma_j, ..., \sigma_n)$
  $\Rightarrow q(\sigma, \sigma') = \begin{cases} \frac{1}{m} & \text{if } \sigma \mapsto \sigma' \text{ is 1 flip} \\ 0 & \text{otherwise} \end{cases}$

- We can show that this q gives a Metropolis chain which is regular (Hint: any 2 states in S are at most m flips apart, $P^m > 0$)

The Gibbs Sampler
ex: Ising model (Glauber dynamics)

$$\begin{aligned}
\text{States} : S &= \{\sigma = (\sigma_1, ..., \sigma_m), \sigma_j = \pm 1\} \\
\text{Target dist} : \pi, \pi(\sigma) &= \frac{b(\sigma)}{z} > 0\ \forall \sigma \in S \\
\text{Notation} : \sigma^{j,+1} &= (\sigma_1, \sigma_2, ..., \sigma_{j-1}, +1, \sigma_{j+1}, ..., \sigma_m) \\
\sigma^{j,-1} &= (\sigma_1, \sigma_2, ..., \sigma_{j-1}, -1, \sigma_{j+1}, ..., \sigma_m)
\end{aligned}$$

Glauber dynamics
(Random-scan Gibbs sampler for Ising model)
Goal: Construct MC, $X_0, X_1, ... \in S$ s.t. $X_n \sim \pi$ as $n \to \infty$
Assume $X_k = \sigma \in S$

1. Choose J uniformly at random from 1,2,..., m
2. Set $\alpha_+ = \frac{\pi(\sigma^{J,+1})}{\pi(\sigma^{J,-1}+\sigma^{J,+1})}$ and $\alpha_- = \frac{\pi(\sigma^{J,-1})}{\pi(\sigma^{J,-1}+\sigma^{J,+1})}$

Note that $\alpha_+ + \alpha_- = 1$
Set $X_{k+1} \leftarrow \sigma^{J,+1}$, with probability α_+
$X_{k+1} \leftarrow \sigma^{J,-1}$, with probability α_-

<u>Remarks</u>

- Each 'spin' is in $\{-1, +1\}$
- $\{\alpha_+, \alpha_-\}$ is pmf over $\{-1, +1\}$
- Set J'th spin conditional on the value of every other spin.

$\boxed{ex}$ $m = 2$ spins

$$\text{States} : S = \{(-1,-1), (-1,+1), (+1,-1), (+1,+1),\}$$
$$\text{Set} : \pi(-1,-1) = \frac{1}{z}, \pi(-1,+1) = \frac{2}{z}$$
$$\pi(+1,-1) = \frac{3}{z}, \pi(+1,+1) = \frac{5}{z}$$

(Here $z = 11$)

Suppose $X_k = (-1,-1), J = 2$ (Glauber dynamics)
Then $\sigma^{J,+1} = (-1,+1)$, $\sigma^{J,-1} = (-1,-1)$
$\alpha_+ = \frac{\frac{3}{z}}{\frac{1}{z}+\frac{3}{z}} = \frac{3}{4}$, $\alpha_- = \frac{1}{1+3} = \frac{1}{4}$
Then $\mathbb{P}(X_{k+1} = (-1,+1) \mid X_k = (-1,-1)) = \frac{3}{4} \cdot \frac{1}{2} = \alpha_+ \cdot \mathbb{P}(J = 2)$

2.7 Lecture 15 - Midterm Review

<u>Recap:</u>

- Inversion, transformation
- Acceptance-rejection method
- MC integration, crude and importance sampling.
- Variance reduction.

- Markov chains (not MCMC)

MC Integration
Goal: Calculate

$$I = \int_D g(x) f(x) dx \quad D \subseteq \mathbb{R}^d$$

$$\begin{aligned} \text{where } D &\text{ is bounded} \\ f, g &: D \to \mathbb{R} \\ f &\text{ is a pdf on } D \end{aligned}$$

For $X \sim f$:

$$\begin{aligned} I &= \mathbb{E}_f(g(x)) \\ \text{Var}_f(g(x)) &= \mathbb{E}_f(g(x)^2) - [\mathbb{E}_f(g(x))]^2 \\ &= \int_D g(x)^2 f(x) dx - I^2 \end{aligned}$$

Crude MC estimator:

$$I_N = \frac{1}{N} \sum_{i=1}^{N} g(x_i) \qquad\qquad X_1, ..., X_N \sim f, iid$$

Then (check):

- $\mathbb{E}(I_N) = I$ (unbiased)
- $\text{Var} I_N = \frac{1}{N} \text{Var}_f(g(x_i))$
- $I_N \to I$ with probability 1

Importance sampling:
The importance function ϕ is a pdf on D.
Then

$$\begin{aligned} I &= \int_D \frac{g(x) f(x)}{\phi(x)} \phi(x) dx \\ &= \mathbb{E}_\phi \left[\frac{g(y) f(y)}{\phi(y)} \right], \quad y \sim \phi \end{aligned}$$

and we have our estimator

$$J_N = \frac{1}{N}\sum_{i=1}^{N}\frac{g(y_i)f(y_i)}{\phi(y_i)} \qquad\qquad y_1, ..., y_N \sim \phi iid$$

Then (check):

- $\mathbb{E}(J_N) = I$ (unbiased)
- $\mathrm{Var}J_N = \frac{1}{N}\left[\int_D \frac{g(x)^2 f(x)^2}{\phi(x)}dx - I^2\right]$
- Choose ϕ to be easy to compute and to give small $\mathrm{Var}J_N$

Variance Reduction
Let $U, V \sim U[0,1],\ \alpha, \beta : \mathbb{R} \to \mathbb{R}$
Set $W = \alpha(U) + \beta(V),\ W$ is an estimator for $\mu = \mathbb{E}(\alpha(U) + \beta(V)) = \mathbb{E}W$
<u>Goal:</u> Find relationship between U, V to reduce Var W
Note: Var $W = \mathrm{Var}(\alpha(U)) + \mathrm{Var}\beta(V) + 2\mathrm{Cov}(\alpha(U), \beta(V))$
Note: $\mathrm{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$ which $= 0$ if X, Y are independent.

1. α, β increasing
   $\to$ Use antithetic RVs: $V = 1 - U$
   $\mathrm{Cov}(\alpha(U), \beta(1-U)) \le 0, \quad \alpha, \beta \uparrow$
2. $\alpha \uparrow, \beta \downarrow$
   $\to$ Use common RVs: $V = U$
   $\mathrm{Cov}(\alpha(U), \beta(U)) \le 0, \quad \alpha \uparrow, \beta \downarrow$
   See practice midterm Q4, HW2 Q4.

Markov Chain

- Discrete time/state, time homogeneous MC
  $X_0, X_1, ... \in S = \{1, 2, 3, ..., N\}$
- Transition matrix
  $P = [p(i,j)],\ p(i,j) = \mathbb{P}(X_1 = j | X_0 = i)\ \forall i, j \in S$
- Distribution vectors: $\pi_n = [\mathbb{P}(X_n = i)]\ \forall i \in S$, pmf X_n
- $\pi_{n+1} = \pi_n P$ (HW3)

- Stationary/invariant distribution
  π s.t. $\pi P = \pi$

- Regular MC or transition matrix means that $\exists\ k \geq 1$ s.t. $P^k > 0$ (Lecture 10, 11)
  Regular $\Rightarrow$ unique stationary distribution $\pi > 0$ and $\pi_n \to \pi$ as $n \to \infty$

$\boxed{ex}$ $P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$ Note that this matrix is regular, with a unique stationary distribution $\Rightarrow \pi_0 P^n \to \pi$ for any π_0.
Solve $\pi P = \pi$, let $\pi = (a, b)$ (HW3, practice midterm)

$$\begin{aligned} \Rightarrow \frac{1}{2}a + \frac{1}{3}b &= a \\ \frac{1}{2}a + \frac{2}{3}b &= b \\ a + b &= 1 \end{aligned}$$

Solving this yields $a = \frac{2}{5}, b = \frac{3}{5}$

Acceptance-Rejection
We have

$$\begin{aligned} f &: \text{target pdf of } Z \in \mathbb{R}^d \\ \alpha g &: \text{proposed pdf, } \ \alpha > 0 \text{ s.t. } \ f(x) \leq \alpha g(x)\ \forall x \end{aligned}$$

<u>Algorithm:</u> sampling $Z \sim f$

1. Generate $X \sim g$
2. Generate $U \sim U[0, 1]$
3. Set $Y = \alpha g(x) \cdot U$
4. If $U \leq \frac{f(x)}{\alpha g(x)}$, then accept X
5. Return $Z \leftarrow X$

<u>Efficiency:</u> $\frac{1}{\alpha} = \frac{\text{vol B}}{\text{Vol A}} =$ acceptance rate

2.8 Lecture 17 - MCMC & Gibbs

Recap - MCMC

- Goal: sample from target distribution f on a state space S, by constructing MC $X_0, X_1, ... \in S$ with $X_n \sim f$ as $n \to \infty$
- Output: $X_0, X_1, X_2, ... \in S$
- For large enough n, $X_n, X_{n+1}, X_{n+2}, ...$ is approximately a sample from f, but $X_n, X_{n+1}, X_{n+2}, ...$ are dependent, not iid.
- For independent sample, we would need to run MCMC many times.
- Estimator: Suppose $X \sim f$ and $Y = g(X)$, we can use MCMC to estimate $\mathbb{E}Y$

  $$\text{Let } \; Y_n = \frac{1}{n}\sum_{i=1}^{n} g(X_i)$$

  Typically, $Y_n \to \mathbb{E}Y$ (HW4)
- Y_n can be used to assess convergence (often)
- Can also assess convergence with sample autocovariance. (HW4)

Gibbs Sampler

- Goal: sample a target pdf

  $$f(x), \; x \in \mathbb{R}^n$$

- Conditional distribution:

  $$f_j(y|x_1, ..., x_n) = \frac{f(x_1, ..., x_{j-1}, y, x_j, ..., x_{m+1})}{\int dy f(x_1, ..., x_{j-1}, y, x_j, ..., x_{m+1})}$$

  for $1 \leq j \leq m$

So $f_j(\, \cdot \mid x_1, ..., x_{m-1})$ is a pdf on $\mathbb{R}$

Algorithm: Random-scan Gibbs Sampler
Given $X_k = (X_{k,1}, ..., X_{k,m}) \in \mathbb{R}^m$

1. Choose J in $\{1, ..., m\}$ randomly (e.g. from uniform distribution on $\{1, ..., m\}$)

2. Sample Y from distribution:

$$f_J(\,\cdot\mid \underbrace{X_{k,1}, ..., X_{k,J-1}, X_{k,J+1}, ..., X_{k,m}}_{\in\mathbb{R}^{m-1}})$$

   Set:

$$X_{k+1} \leftarrow (\underbrace{X_{k,1}, ..., X_{k,J-1}, Y, X_{k,J+1}, ..., X_{k,m}}_{\in\mathbb{R}^{m}})$$

$\boxed{ex}$
Target: $f(x,y) = c\exp(-(x^2+1)(y^2+1)),\ (x,y)\in\mathbb{R}^2$
c is normalization so

$$\int_{\mathbb{R}^2} c\exp(-(x^2+1)(y^2+1))dxdy = 1$$

What are the conditional distributions?

$$\begin{aligned} f_1(x|y) &= \frac{1}{\int_{\mathbb{R}} f(x,y)dx}\cdot c\exp(-(x^2+1)(y^2+1)) \\ &= c_1(y)\cdot\exp(-(y^2+1)x^2) \end{aligned}$$

where $\int_{\mathbb{R}} f(x,y)dx$ is a function of y and $c_1(y)$ is such that

$$\int_{-\infty}^{+\infty} c_1(y)\exp(-(y^2+1)x^2)dx = 1$$

i.e.: $c_1(y)$ is the normalization of the pdf

$$f_1(x|y) \propto \exp(-(y^2+1)x^2)$$

Recall that $W\sim N(0,\sigma^2)$ has pdf $f_w(w)\propto\exp(\frac{-1}{2\sigma^2}w^2)$

$$\Rightarrow f_1(x|y) \text{ is the pdf of } N\left(0, \frac{1}{2(y^2+1)}\right)$$

Similarly, $f_2(y|x)$ is pdf of $N(0,\frac{1}{2(x^2+1)})$
<u>Algorithm</u> (Random-scan Gibbs)
Given $(x_k, y_k)\in\mathbb{R}^2$

1. Choose J uniformly at random from $\{1, 2\}$

2. Generate $Z \sim N(0, 1)$
   If $J = 1$
   Set $W = \frac{1}{\sqrt{2(y_k^2+1)}} z$ (i.e. $W \sim f_1(\cdot \mid y_k)$)
   Set $(x_{k+1}, y_{k+1}) \leftarrow (w, y_k)$
   If $J = 2$ Set $W = \frac{1}{\sqrt{2(x_k^2+1)}} z$ (i.e. $W \sim f_2(\cdot \mid x_k)$)
   Set $(x_{k+1}, y_{k+1}) \leftarrow (x_k, w)$

3. Output: $(x_1, y_1), (x_2, y_2), ...$

Systemic-scan Gibbs Sampler

- Given: $X_k = (X_{k,1}, ..., X_{k,m}) \in \mathbb{R}^m$
- Input: target pdf $f(x)$, $x \in \mathbb{R}^m$

1. Sample Y_1 from $f_1(\cdot \mid X_{k,2}, ..., X_{k,m})$
   For $i = 2$ to m
   Sample Y_i from

$$f_i(\cdot \mid y_1, ..., y_{i-1}, x_k)$$

2. Set $X_{k+1} \leftarrow (Y_1, \ldots, Y_m)$

3. Output: $X_0, X_1, \cdots \in \mathbb{R}^m$

Remarks:

- Systemic-scan $\rightarrow$ choose coords $J = 1, 2, \ldots, m$ in order
- Random-scan Gibbs is a special case of Metropolis.

Metropolis (Random-Walk Sampler)
Goal: sample from target pdf $f(x)$, $x \in \mathbb{R}^m$, where $f(x)$ has an unknown normalization.
Given $X_k = (X_{k1}, \ldots, X_{km}) \in \mathbb{R}^m$

1. Generate $Z = (Z_1, \ldots, Z_m) \in \mathbb{R}^m$
   Where $Z_1, \ldots, Z_m \sim N(0, 1)$ iid
   (i.e. $Z \sim N(0, I_m)$ is a multivariate normal)

2. Set proposal state $Y = X_k + Z$
   (proposal 'transition matrix' $q(X_k, \cdot)$ is a pdf of $N(X_k, I_m)$)

3. Set $\alpha = \min(1, \frac{f(Y)}{f(X_k)})$

4. Generate $U \sim U[0,1]$
   Set $X_{k+1} \leftarrow Y$ if $U \leq \alpha$
   Set $X_{k+1} \leftarrow X_k$ if $U > \alpha$

5. Output: $X_0, X_1, \dots \in \mathbb{R}^m$

Remarks:

- Output is a MC on a continuous state space $\mathbb{R}^m$
- Sensitive to typical 'jump' size Z (HW 4)
- What happens if we take $\updownarrow Z$ and let $\updownarrow \to 0$ and we look at X_n as $n \to \infty$?
  $\to$ SDEs

3 Stochastic Differential Equation

3.1 Lecture 18 - Intro to SDEs

- Continuous-time, continuous state space.
- Markov processes, e.g. $X(t) \in \mathbb{R},\ t \in \mathbb{R}$
- Key example: Brownian motion
- Goals: Simulate $X(t)$, sample from the distribution $X(t)$, find distribution of $X(t)$, compute e.g. moments, $\mathbb{E}(X(t)), \mathbb{E}(X(t)^2)$
- Learn numerical & analytical approaches for these computations.

Preliminaries: Non-stochastic processes
ODE example:

$$\text{Let } x(t) \in \mathbb{R},\ t \in [0, \infty)$$
$$\text{Consider } \frac{dx}{dt} = x,\ x(0) = 1$$
$$\Rightarrow \text{Solution: } x(t) = Ae^t$$
$$\text{Initial condition } \Rightarrow x(t) = e^t$$

Discretization
We can simulate/solve the ODE by discretizing it.

$$\text{Let } \delta t > 0 \text{ be small and}$$
$$\text{Let } t_n = n \cdot \delta t,\ n = 0, 1, 2, ...$$

How are $x(t_n), x(t_{n+1})$ related?

$$\text{We have } x(t_n) = \frac{dx}{dt}\Big|_{t=t_n} \approx \frac{x(t_{n+1}) - x(t_n)}{\delta t}$$
$$\Rightarrow x(t_{n+1}) \approx x(t_n) + x(t_n) \cdot \delta t$$

Consider the difference equation:

$$x_{n+1} = x_n + x_n \cdot \delta t,\ x_0 = 1,\ n = 0, 1, 2, ...$$
$$x_1 = 1 + \delta t,\ x_2 = (1 + \delta t)^2,\ ...\ ,\ x_n = (1 + \delta t)^n = (1 + \frac{t_n}{n})^n$$

- Does the discrete solution x_n approach the continuous solution $x(t)$?

Let $t' = t_n = n\delta t$ and consider $\lim_{n\to\infty} \delta t \to 0$ with t' fixed.
Then $x_n = (1 + \frac{t'}{n})^n \to e^{t'} = x(t')$ as $n \to \infty,\ \delta t \to 0,\ t' = n\delta t$ fixed.

Remark:

- We can think of an ODE $\frac{dx}{dt} = x$ as a difference equation $x_{n+1} = x_n + x_n\delta t$ in the limit $\delta \to 0$

Continuous limit of difference equation
Consider difference equation $x_n \in \mathbb{R}$

$$x_{n+1} = x_n + \delta t \cdot u,\ x_0 = 0,\ n = 1, 2, ...$$

Is there a continuum limit as $\delta t \to 0$? $\Leftrightarrow$ Is there an $x(t)$ such that

$$x_n \to x(t) \text{ as } n \to \infty$$
$$\delta t \to 0$$
$$\text{with } n\delta t = t \text{ fixed}$$

Check:

$$x_1 = \delta t u, \ \dots, \ x_n = n \delta t u$$
$$= tu \to t \cdot u$$
$$\Rightarrow \text{Continuum limit } x(t) = tu, \ \frac{dx}{dt} = u$$

<u>Remark</u>: Suppose we took

$$x_{n+1} = x_n + \delta t^{\alpha} u$$

Then $x_n = n \cdot \delta t^{\alpha} \cdot u = t \cdot \delta t^{\alpha - 1} \cdot u$

$$x_n \to \begin{cases} 0 & \text{if } \alpha > 1 \\ u \cdot t & \text{if } \alpha = 1 \\ \infty & \text{if } \alpha < 1 \end{cases} \text{ as } \begin{cases} \delta t \to 0 \\ n \to \infty \\ n\delta t = t \text{ fixed} \end{cases}$$

Sensible continuum limit only for $\alpha = 1$

<u>Stochastic processes</u>

Discrete time Continuous space	$\to$	Continuous time Continuous space

- Random walk on $\mathbb{R}$
  Let $Z_0, Z_1, ..., Z_n \sim N(0,1)$ *iid* and consider $X_{n+1} = X_n + \delta t^{\alpha} Z_n, \ n = 0, 1, 2, ...$ with $X_0 = 0$

- $X_1 = \delta t^{\alpha} \cdot Z_0, \ X_2 = \delta t^{\alpha}(Z_0 + Z_1)$

$$X_n = \sum_{i=0}^{n-1} \delta t^{\alpha} Z_i$$

- Sum of normals is normals
  $\Rightarrow \mathbb{E}X_n = 0,\ \mathrm{Var}X_n = n \cdot \delta t^{2\alpha}$
  $\Rightarrow X_n \sim N(0, n\delta t^{2\alpha})$

- Continuum limit: is there $X(t)$ such that $X_n \to X(t)$ as $n \to \infty$, $\delta t \to 0$, with $n\delta t = t$ fixed.
  We have:

$$\mathrm{Var}X_n = n\delta t^{2\alpha} = t \cdot \delta t^{2\alpha-1} \to \begin{cases} 0 & \alpha > \frac{1}{2} \\ \infty & \alpha < \frac{1}{2} \\ t & \alpha = \frac{1}{2} \end{cases}$$

$$\text{as } n \to \infty, \delta t \to 0,\ \text{with } n\delta t = t \text{ fixed.}$$

  Sensible continuum limit only for $\alpha = \frac{1}{2}$

3.2 Lecture 19 - Brownian Motion

Goals:

- Define Brownian motion
- SDEs
- Simulation

Recap:

- Discrete-time, random walk on $\mathbb{R}$
  Let $Z_0, Z_1, ..., \sim N(0,1)$ *iid*
  and consider

$$X_{n+1} = X_n + \delta t^{\alpha} Z_n,\ n = 0, 1, 2, \dots$$
$$X_0 = 0,\ \delta t > 0$$
$$\Rightarrow X_n = \delta t \sum_{i=0}^{n-1} Z_i \sim N(0, n \cdot \delta t^{2\alpha})$$

Take limit $n \to \infty,\ \delta t \to \infty,\ n \cdot \delta t = t$ fixed.

$$\begin{aligned}\mathrm{Var} X_n &= n \cdot \delta t^{2\alpha} \\ &= t \cdot \delta t^{2\alpha - 1} \begin{cases} 0 & \alpha > \frac{1}{2} \\ \infty & \alpha < \frac{1}{2} \\ t & \alpha = \frac{1}{2} \end{cases}\end{aligned}$$

- For $\alpha = \frac{1}{2}$, there is finite, non-zero $X(t)$ s.t. $X_n \to X(t)$ as $n \to \infty,\ \delta t \to \infty,\ n \cdot \delta t = t$ fixed.

Remarks:

- Above discussion is non-rigorous
- Continuum limit $X(t)$ is called Brownian motion (BM) / Wiener process.
- Discretized BM

$$\begin{aligned}&X_{n+1} = X_n + \sqrt{\delta t} \cdot Z_n,\ X_0 = 0 \\ &\text{for } Z_0, Z_1, \ldots, \sim N(0,1)\ iid,\ \delta t > 0\end{aligned}$$

- SDE notation for $X(t) \in \mathbb{R}$

$$dx = dB$$

Infinitesimal increment of X = Infinitesimal increment of BM

(Here $X(t)$ is a BM)

- We can think of SDEs in terms of the discrete equation, with $\delta t \to 0$ (informal)

Brownian Motion (BM).

Def: A stochastic process $B(t) \in \mathbb{R},\ t \geq 0$ is a BM if:

1. $B(0) = 0$ with probability 1.
2. $B(t) - B(s) \sim N(0, t-s)\ \forall\ t \geq s \geq 0$
   $[B(t) \cdot B(s) = \sqrt{t-s} \cdot Z$ for $Z \sim N(0,1)]$
   $[i.e.\ B(t) \sim N(0,t)]$

3. For any $0 \leq t_0 \leq t_1 \leq ... \leq t_n$,
   the RVs $B(t_1) - B(t_0), ... , B(t_n) - B(t_{n-1})$ are independent (independent increments).

4. $B(t)$ is a continuous function of $t \geq 0$

Remarks:

- From 2. $B(t + \delta t) - B(t) \sim N(0, \delta t),\ \ \delta t > 0$
  $\Rightarrow$ Have $B(t + \delta t) - B(1) = \sqrt{\delta t} \cdot Z$ for $Z \sim N(0, 1)$

- BM is continuous, but nowhere differentiable
  From above:

$$\begin{aligned} \frac{B(t + \delta t) - B(t)}{\delta t} &= \frac{\sqrt{\delta t} \cdot Z}{\delta t},\ \text{ for } Z \sim N(0,1) \\ &= \frac{1}{\sqrt{\delta t}} Z \end{aligned}$$

Properties of BM
$B(t),\ t \geq 0$ is a Markov process.
For $0 \leq t_1 \leq ... \leq t_n \leq t$, we have

$$\mathbb{P}(B(t) \leq b \mid B(t_1), \dots, B(t_n)) = \mathbb{P}(B(t) \leq b \mid B(t_n))$$

$\boxed{ex}$ The following processes are BMs:

- $\tilde{B}(t) = -B(t), t \geq 0$
- $\tilde{B}(t) = B(t + t_0) - B(t_0), t \geq 0$
- $\tilde{B}(t) = \frac{1}{\sqrt{\lambda}} B(\lambda t), \lambda > 0$
- (e.g. check $\tilde{B}(t)$ satisfies 4 properties if $B(t)$ does)

Transition probability density: $p(y, t \mid x, s)$ for $s \leq t$

$$\begin{aligned} \int_a^b p(y, t \mid x, s) dy &= \mathbb{P}(a < B(t) < b \mid B(s) = x) \\ &= \mathbb{P}(a - x < B(t) - B(s) < b - x) \\ &= \mathbb{P}(a < \sqrt{t \cdot s} \cdot Z + x < b) \text{ for } Z \sim N(0, 1) \end{aligned}$$

$$\Rightarrow p(y, t \mid x, s) = \frac{1}{\sqrt{2\pi(t \cdot s)}} \exp(-\frac{(y - x)^2}{2(t - s)^2})$$

$B(t), B(s)$ are not independent.

$$\begin{aligned}\mathrm{Cov}(B(t), B(s)) &= \mathbb{E}(B(t) \cdot B(s)) - \mathbb{E}(B(t))\mathbb{E}(B(s)) \\ &= \mathbb{E}(B(t) \cdot B(s)) - 0 \text{ because } B(t) \sim N(0,t) \\ &= \mathbb{E}(B(t)B(s))\end{aligned}$$

Assume $t \geq s$, then

$$\begin{aligned}B(t)B(s) &= [B(t) - B(s)][B(s) - B(0)] + B(s)^2 \\ \mathbb{E}(B(t)B(s)) &= \mathbb{E}(B(t) - B(s))\mathbb{E}(B(s)B(0)) + \mathbb{E}(B(s)^2) \\ &= s\end{aligned}$$

$$\Rightarrow \mathbb{E}(B(s)B(t)) = \min(s,t)$$

3.3 Lecture 20 - SDEs

Recap:
Definition of Brownian motion, $B(t) \in \mathbb{R}, t \geq 0$

1. $B(0) = 0$, with probability 1
2. $B(t) - B(s) \sim N(0, t-s)$ for $t \geq s$
3. Independent increments
   $\forall\ 0 \leq s_1 < t_1 \leq s_2 < t_2$
   RVs $B(t_1) - B(s_1), B(t_2) - B(s_2)$ are independent.
4. $B(t)$ is a continuous function of t (but not differentiable).

1. and 2. $\Rightarrow B(t) \sim N(0,t)$
Discrete picture: $X_0, X_1, \dots \in \mathbb{R}, \delta t > 0$

$$\begin{aligned}X_{n+1} = X_n + \sqrt{\delta t} \cdot Z_n,\ n = 0, 1, 2, \dots \\ Z_0, Z_1, \dots \sim N(0,1) \text{ iid}, X_0 = 0\end{aligned}$$

Continuum limit: $n \to \infty,\ \delta t \to 0, n \cdot \delta t = t$ fixed.

$$X_n \to X(t),\ X(t) \text{ is a BM}$$

Simulating BM (Euler-Maruyama)
Let $\delta t > 0$, $n \in \mathbb{Z}$, $T = n\delta t$

Let $t_k = k\delta t,\ k = 0, 1, 2, \dots, n,\ t_n = T = n\delta t$
Idea: Find a discrete time process on $[0, T]$

$$X(t_k) = X_k \text{ with } X_k \approx B(t_k),\ \text{for BM } B(t)$$

Algorithm: (Euler)

1. Set $X_0 = 0$
2. For $i = 0$ to $n - 1$
3. Generate $Z \sim N(0, 1)$
4. Set $X_{i+1} = X_i + \sqrt{\delta t} \cdot Z$
5. Output: $X_0, X_1, \dots, X_n = X(0), X(t_1), \dots, X(t_n)$

Remarks:

- Can interpolate to connect $(t_k, X(t_k))$ to get approximate sample path $X(t)$ on $[0, T]$ with $X(t) \approx B(t)$, BM
- Return to convergence later.

BM with Drift

$$X(t) = ut + \sqrt{2D}B(t)$$

where $u, D \in \mathbb{R},\ D > 0$ where $B(t)$ is a BM

- Distribution of $X(t)$
  $X(t) \sim$ normal (constant + normal)
- $\mathbb{E}X(t) = ut$
- $\text{Var}X(t) = \mathbb{E}((X(t) - ut)^2)$
  $= \mathbb{E}(2D)B(t)^2$
  $= 2Dt(\mathbb{E}(B(t)^2) = t)$
- $\Rightarrow X(t) \sim N(ut, 2Dt)$

<u>SDE notation</u>: SDE whose solution is BM with drift

$$\underbrace{dx}_{\text{change in X(t)}} = \underbrace{udt}_{\text{change in t}} + \sqrt{2D} \cdot \underbrace{dB}_{\text{infinitesimal Brownian increment}}, \; X(0) = 0$$

Here you can 'integrate' and get the original equation $X(t) = ut + \sqrt{2D}B(t)$. However, stochastic integration is different in general.

<u>Euler's method</u>:
Let $\delta t > 0,\ n \in \mathbb{Z},\ t_k = k\delta t,\ k = 0, 1, \ldots, n$
Set $X_0 = 0$
For $i = 0$ to $n - 1$
  Generate $Z \sim N(0, 1)$
  Set $X_{i+1} = X_i + u\delta t + \sqrt{2D}\sqrt{\delta t}Z$
End

<u>Output</u>: $X_0, X_1, \ldots, X_n$
X_i approximate true solution $X(t_i)$

<u>Remarks</u>:

- SDE: $dx = udt + \sqrt{2D}dB$
- EM: $X_{i+1} = X_i + u\delta t + \sqrt{2D\delta t}Z_i$
  $Z_0, Z_1, \cdots \sim N(0, 1)$ iid
- Can think of SDE solution $X(t)$ as 'continuum limit' of solution to discrete equation.
- Intuitively, dB is like a $\sqrt{dt}Z$
  $dB(t) = B(t + dt) - B(t) \sim N(0, dt)$
  $\mathbb{E}(dB) = 0,\ \mathbb{E}(dB^2) = dt$
- Langevin notation (physics)
  If we divide by dt

$$\frac{dx}{dt} = u + \sqrt{2D} \underbrace{\xi(t)}_{\text{white noise } \frac{dB}{dt}}$$

Note that $\xi(t)$ is not a function!
From above $\frac{dB}{dt} \sim \frac{1}{\sqrt{dt}} z$

$\boxed{\text{ex}}$ Moments of BM with drift

$$\begin{aligned}
dX &= udt + \sqrt{2D} dB \\
\mathbb{E}(dX) &= udt \\
d(\mathbb{E}X) &= udt \text{ (ODE for } \mathbb{E}X) \\
\frac{d}{dt}(\mathbb{E}X) &= u \\
\mathbb{E}X &= ut
\end{aligned}$$

From the discrete picture:

$$\begin{aligned}
X_{i+1} - X_i &= u\delta t + \sqrt{2D\delta t} Z \\
\mathbb{E}X_{i+1} - \mathbb{E}X_i &= u\delta t \\
\frac{\mathbb{E}X_{i+1} - \mathbb{E}X_i}{\delta t} &= u \\
\frac{d}{dt}\mathbb{E}X &= u \\
\mathbb{E}X &= ut
\end{aligned}$$

3.4 Lecture 21 - General SDEs

Recap:

- BM with drift: $X(t) \in \mathbb{R}$
- SDE: $dX = udt + \sqrt{2D} \cdot dB, X(0) = 0$
  Where dX is change in X, dt is change in t, dB is the infinitesimal Brownian increment ($dB \sim N(0, dt)$). Where $u, D \in \mathbb{R}, D > 0$, and u is speed, D is diffusivity. Intuitively, dB is like $\sqrt{dt} \cdot Z,\ Z \sim N(0, 1)$, and $\mathbb{E}(dB) = 0,\ \mathbb{E}(dB^2) = dt$.

Why does $\mathbb{E}(dx) = d(\mathbb{E}X)$?

Let $y(t) = \mathbb{E}(X(t))$, then

$$\begin{aligned}
\mathbb{E}(dx) &= \mathbb{E}(X(t+\delta t) - X(t)) \\
&= \mathbb{E}(X(t+\delta t)) - \mathbb{E}X(t) \\
&= y(t+\delta t) - y(t) \\
&= dy \\
&= d(\mathbb{E}X)
\end{aligned}$$

Ornstein-Uhlenbeck process (OU)

$$dX = -\alpha X dt + \sqrt{2D} dB, \ X(0) = x_0$$

Constants: $\alpha, D, x_0 \in \mathbb{R}, D > 0$
1d process: $X(t) \in \mathbb{R}$
Euler-Maruyama: numerical approximation for $X(t)$ on $t \in [0, T]$
Algorithm:

1. Set $T > 0, \ n \in \mathbb{Z}_{>0}, \ \delta t = \frac{T}{n}$
2. Let $t_k = k\delta t, \ k = 0, 1, \ldots, n$
3. For $i = 0, \ldots, n-1$
   Generate $Z \sim N(0,1)$
   Set $X_{i+1} = X_i - \alpha X_i \delta t + \sqrt{2D\delta t} \cdot Z$
4. End

Output: $X_0, X_1, \ldots, X_n$ with $X_i \approx X(t_i)$ (true solution).

What are the moments of $X(t)$?

$$\begin{aligned}
dX &= -\alpha X dt + \sqrt{2D} dB \\
d(\mathbb{E}X) &= -\alpha \mathbb{E}X dt \\
\frac{d}{dt}\mathbb{E}X &= -\alpha \mathbb{E}X
\end{aligned}$$

This gives an ODE for $\mathbb{E}X$

Let $y(t) = \mathbb{E}(X(t))$

$$\begin{aligned}
&\Rightarrow \frac{dy}{dt} = -\alpha y, \ y(0) = x_0 \\
&\Rightarrow y(t) = x_0 e^{-\alpha t} = \mathbb{E}(X(t))
\end{aligned}$$

Remarks:

- What about higher moments, e.g. $\mathbb{E}(X^2)$?
- What is $d(\mathbb{E}(X^2)) = \mathbb{E}(d(X^2))$?
  Ito's lemma, stochastic calculus.

General SDEs In 1d, SDEs are equations of the form:

$$dX = b(X,t)dt + \sigma(X,t)dB$$

Initial condition: $X(0) = x_0$
Where: $B(t)$ is a BM, $X(t) \in \mathbb{R}$ (1d SDE), $b(X,t), \sigma(X,t) \in \mathbb{R}$
Terminology:

- Drift term: $b(X,t)dt$
- Diffusion term: $\sigma(X,t)dB$
- Time homogeneous: b, σ are independent of t, it would be time inhomogeneous otherwise
- Additive noise: σ is independent of X
- Multiplicative noise: σ depends on X

$\boxed{\text{ex}}$ Particle in force field $F(X)$, $X(t) \in \mathbb{R}$

$$\text{SDE}: \quad dX = \underbrace{F(X)dt}_{\text{drift, deterministic force}} + \underbrace{\sqrt{2D}dB}_{\text{diffusion, random kicks}}$$

- $D > 0$ is constant.
- SDE is time homogeneous with additive noise.

Note:
SDEs are sometimes written as

$$\begin{aligned} dX_t &= b(X_t,t)dt + \sigma(X_t,t)dB_t \\ \frac{dX}{dt} &= b(X,t) + \sigma(X,t)\xi \\ &\text{(Langevin)} \end{aligned}$$

SDE in higher dimensions

- $B(t) = (B_1(t), \ldots, B_m(t)) \in \mathbb{R}^m$ is BM if all components are independent BMs
- SDE for $X(t) \in \mathbb{R}^m$ has the form

$$dX = b(X,t)dt + \sigma(X,t)dB$$

  Where $b(X,t) \in \mathbb{R}^m$ and in general, $\sigma(X,t)$ can be $m \times m$ matrix (if $\sigma \propto I_m$, then can treat σ as scalar)

$\boxed{\text{ex}}$ Brownian particle in 2d, $X(t) \in \mathbb{R}^2$

$$\text{SDE: } dX = \sqrt{2D}dB$$

Euler: $T,\ \delta t > 0,\ n = \frac{T}{\delta t},\ t_k = k\delta t,\ k = 0, \ldots, n$
For $i = 0, \ldots, n-1$
  Generate $Z_1, Z_2 \sim N(0,1)$ iid
  $X_{i+1} = X_i + \sqrt{\delta t}\begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix}$
End

$\boxed{\text{ex}}$ Active Brownian particle

- Speed v
- Direction $n_\theta = (\cos\theta, \sin\theta)$
- θ diffuses

$$\text{SDE: } dX = vn_\theta dt + \sqrt{2D_T}dB_T$$
$$d\theta = \sqrt{2D_R}dB_R$$

Where $X(t) \in \mathbb{R}^2,\ \theta(t) \in \mathbb{R}$, and constants $v, D_R, D_T \in \mathbb{R}$

- $B_T(t) \in \mathbb{R}^2$ is 2d BM
- $B_R(t) \in \mathbb{R}$ is 1d BM
- B_T, B_R are independent processes

Euler: Choose $T, \delta t > 0, n = \frac{T}{\delta t},\ x_0, \theta_0$
For $i = 0, \dots, n-1$
Generate $Z_1, Z_2, Z_3 \sim N(0,1)$ iid.
Set:

$$\begin{aligned} X_{i+1} &= X_i + v \begin{bmatrix} \cos\theta_i \\ \sin\theta_i \end{bmatrix} \delta t + \sqrt{2D_T \delta t} \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} \\ \theta_{i+1} &= \theta_i + \sqrt{2D_R \delta t} Z_3 \end{aligned}$$

End
Output $(\theta_i,\ X_i) \approx (\theta(t_i), X(t_i))$

3.5 Lecture 22 - Ito's Lemma & Stochastic Calculus

Recap:

- SDE $X(t) \in \mathbb{R},\ t \geq 0$
  $dX = u(X,t)dt + \sigma(X,t)dB$, where $\sigma(X,t)$ are sometimes $\sqrt{2D(X,t)}$
- $B(t)$ is BM, $u(X,t),\ \sigma(X,t) \in \mathbb{R}$
- Time homogeneous: $u,\ \sigma$ are independent of t
- Additive noise: σ independent of X
- Multiplicative noise: σ depends on X
- SDE for $X(t) \in \mathbb{R}^m$

$$dX = u(X,t)dt + \sigma(X,t)dB$$

- $B(t)$ is a m-dimensional BM
- σ is a $m \times m$ matrix in general.

Question: How do we compute moments, e.g. what is $d(\mathbb{E}(X^2)) = \mathbb{E}(d(X^2))$?

Ito's Lemma Background
Stochastic chain rule for computing $d(f(x))$, where $X(t)$ solves an SDE.

Taylor Series:
Let $f : \mathbb{R} \to \mathbb{R}$ be smooth. $\epsilon > 0$

$$f(a+\epsilon) = f(a) + \underbrace{\epsilon f'(a)}_{\mathcal{O}(\epsilon)} + \underbrace{\frac{1}{2}\epsilon^2 f''(a)}_{\mathcal{O}(\epsilon^2)} + \dots$$

A function $g(\epsilon)$ is $\mathcal{O}(\epsilon^k)$ as $\epsilon \to 0$ if $\frac{g(\epsilon)}{\epsilon^k}$ is bounded as $\epsilon \to 0$

$\boxed{\text{ex}}$

- $2\epsilon = \mathcal{O}(\epsilon)$ since $\frac{2\epsilon}{\epsilon} = 2$ bounded as $\epsilon \to 0$
- $2\epsilon \neq \mathcal{O}(\epsilon^2)$ since $\frac{2\epsilon}{\epsilon^2} = \frac{2}{\epsilon}$ not bounded as $\epsilon \to 0$
- $\epsilon + \frac{1}{2}\epsilon^2 = \mathcal{O}(\epsilon)$ as $\epsilon \to 0$
- $e^x = 1 + x + \frac{1}{2}x^2 + \mathcal{O}(x^3)$ as $x \to 0$
  $e^x - (1 + x + \frac{x^2}{2}) = \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \dots$
- $f(x+\epsilon) = f(x) + \epsilon f'(x) + \mathcal{O}(\epsilon^2)$

Consider a 1d, time homogeneous case $X(t) \in \mathbb{R}$

$$dX = \underbrace{u(X)dt}_{\mathcal{O}(dt)} + \underbrace{\sigma(X)dB}_{\mathcal{O}(\sqrt{dt})}$$

Recall:

$$dB \sim \sqrt{dt}Z \sim \mathcal{O}(\sqrt{dt})$$
$$dB \sim N(0, dt)$$

Given function $f(X)$ and dX, find expression for $df = f(X + dX) - f(X)$.
Strategy: Taylor expand f up to $\mathcal{O}(dt)$ and treat $dB^2 = dt$
The intuition for $dB^2 = dt$ is

$$dB^2 = \underbrace{\mathbb{B}^{\nvDash}}_{dt} + \text{ small fluctuations}$$

Treat $dB \sim \sqrt{\delta t}Z,\ Z \sim N(0,1),$ add n copies with $T = n\delta t$ fixed.

$$\sum_{i=1}^{n} \delta t Z_i^2 \sim N(n\delta t, n\delta t^2) \to T$$

Calculate:

$$\begin{aligned}
df &= f(X+dX) - f(X)\\
&= f(X) + f'(X)dX + \frac{1}{2}f''(X)dX^2 + \mathcal{O}(dX^3)\\
&= f'(X)dX + \frac{1}{2}f''(X)dX^2 + \mathcal{O}(dt^{\frac{3}{2}})\\
&= f'(X)(u(X)dt + \sigma(X)dB) + \frac{1}{2}f''(X)((\sigma(X)dB)^2 + \mathcal{O}(dt^{\frac{3}{2}}))\\
&= \underbrace{(u(X)f'(X) + \frac{1}{2}f''(X)\sigma(X)^2)dt + f'(X)\sigma(X)dB}_{\text{Ito's Lemma}}
\end{aligned}$$

Remarks:

- Stochastic chain rule:
  If $\sigma < 0$, then $dX = u(X)dt$ and

$$\begin{aligned}
df &= f(x+dx) - f(x)\\
&= f'(x)u(x)dt\\
&= f'(x)dx
\end{aligned}$$

- Stochastic case:

$$df \neq f'(x)dx \text{ at } \mathcal{O}(dt) \text{ in general because } dX \sim \mathcal{O}(\sqrt{dt})$$

- $dx \sim \mathcal{O}(\sqrt{dt}) \Rightarrow$ needs to work to $\mathcal{O}(dX^2)$ to get an expression correct to $\mathcal{O}(dt)$

$$dX = \underbrace{u(X)dt}_{\mathcal{O}(dt)} + \underbrace{\sigma(x)dB}_{\mathcal{O}(\sqrt{dt})} = \mathcal{O}(\sqrt{dt})$$

- What about the $(\sigma(x)dB)^2$ term?

Discretizations:
ODE case:
Consider a non-stochastic ODE

$$\frac{dx}{dt} = u,\ x(0) = \alpha$$

Let $\delta t > 0$, set $t_i = i\delta t$
Then

$$\begin{aligned}
x(t_{i+1}) &= x(t_i) + \delta t \frac{dx}{dt} + \mathcal{O}(\delta t^2) \\
&= x(t_i) + \delta t \frac{dx}{dt}\Big|_{t=t_i} + \mathcal{O}(\delta t^2) \\
&= x(t_i) + \delta t u(x(t_i)) + \mathcal{O}(\delta t^2) \\
\text{But } u(x(t_i)) &= u(x(t_i) + \mathcal{O}(\delta t)) \\
&= u(x(t_i)) + \mathcal{O}(\delta t) \quad \text{at } \mathcal{O}(\delta t) \\
\Rightarrow x(t_{i+1}) &= x(t_i) + u(x(t_{i+1}))\delta t + \mathcal{O}(\delta t^2) \\
\text{and } x(t_{i+1}) &= x(t_i) + u(x(t_i))\delta t + \mathcal{O}(\delta t^2)
\end{aligned}$$

We have equivalent discretizations of $x(t)$

$$\begin{aligned}
&x_{i+1} = x_i + u(x_i)\delta t && x_0 = a \\
&\tilde{x}_{i+1} = \tilde{x}_i + u(\tilde{x}_{i+1})\delta t && \tilde{x}_0 = a
\end{aligned}$$

where both $x_i, \tilde{x}_i \approx x(t_i)$

<u>SDE case</u>:
Consider an SDE:

$$dX = u(X)dt + \sigma(X)dB,\ X(0) = x_0$$

Set $\delta t > 0$, discretize up to $\mathcal{O}(\delta t)$

$$X_{i+1} = X_i + u(X_i)\delta t + \sigma(X_i)\sqrt{\delta t} Z_i$$

where $X_0 = x_0$ and $Z_i \sim N(0,1)$ and $Z_i,\ X_i$ independent.
This is <u>different</u> from:

$$X_{i+1} = X_i + u(X_{i+1})\delta t + \sigma(X_{i+1})\sqrt{\delta t} Z_i$$
$$Z_i \sim N(0,1) \text{ is independent of } X_i$$

Ito's SDE: Continuum limit of

$$X_{i+1} = X_i + u(X_i)\delta t + \sigma(X_i)\sqrt{\delta t} Z_i$$
$$Z_i \sim N(0,1) \text{ is independent of } X_i$$

3.6 Lecture 23 - Application of Ito's Lemma

Recap:

- Ito's Lemma: stochastic chain rule
- 1d, time-homogeneous: $dX = \underbrace{u(X)dt}_{\mathcal{O}(dt)} + \underbrace{\sigma(X)dB}_{\mathcal{O}(\sqrt{dt})}$

Change in $f(X)$ is

$$\begin{aligned}
df &= f(X+dX) - f(x) \\
&= f'(X)dX + \frac{1}{2}f''(X)dX^2 + \underbrace{\mathcal{O}(dX^3)}_{\mathcal{O}(dt^{\frac{3}{2}})} \\
&= f'(X)[u(X)dt + \sigma dB] + \frac{1}{2}f''(X)\sigma(X)^2\underbrace{dB^2}_{dt} + \mathcal{O}(dt^{\frac{3}{2}}) \\
&= (uf' + \frac{1}{2}\sigma^2 f'')dt + f'\sigma dB
\end{aligned}$$

$\boxed{\text{ex}}$ Ornstein-Uhlenbeck process: $X(t) \in \mathbb{R}$
SDE: $dX = -\alpha X dt + \sqrt{2D}dB,\ X(0) = x_0$
where $\alpha,\ D$ are constants, $\alpha,\ D > 0$

What is df for $f(X) = X^2$?
Ito's:

$$\begin{aligned}
df &= (-\alpha X f' + D f'')dt + f'\sqrt{2D}dB \\
d(X^2) &= (-2\alpha X^2 + 2D)dt + \underbrace{2X\sqrt{2D}dB}_{\text{mult. noise}}
\end{aligned}$$

Discretization, types of SDE

- Multiplicative noise can have multiple meanings. We mostly work in the time homogeneous case.

Ito SDE:
Continuum limit of:

$$\begin{aligned}
X_{i+1} &= X_i + u(X_i)\delta t + \sigma(X_i)\sqrt{\delta t}Z_i \\
Z_i &\sim N(0,1),\ X_i, Z_i \text{ independent},\ \delta t > 0
\end{aligned}$$

SDE: $dX = u(X)dt + \sigma(X)dB \leftarrow$ Most common type of SDE
<u>Remarks</u>:

- $\mathbb{E}(\sigma(X_i)\sqrt{\delta t}Z_i) = 0$ (independent)
  $\Rightarrow \mathbb{E}(\sigma(X)dB) = 0$
  $\Rightarrow \mathbb{E}(dX) = \mathbb{E}(u(X))dt$

<u>Backwards-Ito SDE</u>:
Continuum limit of:

$$X_{i+1} = X_i + u(X_{i+1})\delta t + \sigma(X_{i+1})\sqrt{\delta t}Z_i$$
$$Z_i \sim N(0,1),\ X_i, Z_i \text{ independent},\ \delta t > 0$$

SDE: $dX = u(X)dt + \sigma(X) \bullet dB$
<u>Remarks</u>:

- Anticipatory: diffusion term depends on future state.
- $\mathbb{E}(dX) = \mathbb{E}(u(X))dt$

<u>Stratanovich SDE</u>:
Continuum limit of:

$$X_{i+1} = X_i + u(\frac{X_i + X_{i+1}}{2})\delta t + \sigma(\frac{X_i + X_{i+1}}{2})\sqrt{\delta t}Z_i$$
$$Z_i \sim N(0,1),\ X_i, Z_i \text{ independent}, \delta t > 0$$

SDE: $dX = u(X)dt + \sigma(X) \circ dB$
<u>Remarks</u>:

- Notation can vary
- Ito-SDE sometimes written as

$$dX = u(X)dt + \sigma(X) * dB$$

- We consider Ito SDEs, which is most common since $\mathbb{E}(dX) = \mathbb{E}(u(X))dt$
- Different SDEs can be transformed into each other.

$\boxed{\text{ex}}$ Suppose $X(t)$ satisfies

$$dX = u(X)dt + \sigma(X) \bullet dB \quad \text{(Backwards Ito)}$$

Then $X(t)$ satisfies

$$dX = (u(X) + \sigma(X)\sigma'(X))dt + \sigma(X)dB \quad \text{(Ito)}$$

- Different SDEs have different chain rules, Ito's lemma holds for Ito SDEs.

$\boxed{\text{ex}}$ Using discretizations, find $df(X)$.
Ito SDE: $X_{i+1} = X_i + u(X_i)\delta t + \sigma(X_i)\sqrt{\delta t}Z_i$

$$\begin{aligned}
\Rightarrow f(X_{i+1}) &= f(X_i) + f'(X_i)[u(X_i)\delta t + \sigma(X_i)\sqrt{\delta t}Z_i] \\
&\quad + \frac{1}{2}f''(X_i)\sigma(X_i)^2 \underbrace{\delta t Z_i^2}_{\delta t} + \mathcal{O}(\delta t^{\frac{3}{2}}) \\
\Rightarrow f(X_{i+1}) &= f(X_i) + \delta t[u(X_i)f'(X_i) + \frac{1}{2}\sigma(X_i)^2 f''(X_i)] \\
&\quad + f'(X_i)\sigma(X_i)\sqrt{\delta t}Z_i + \mathcal{O}(\delta t^{\frac{3}{2}})
\end{aligned}$$

$\boxed{\text{ex}}$ OU-process again $X(t) \in \mathbb{R}$
SDE: $dX = -\alpha X dt + \sqrt{2D}dB,\ X(0) = x_0$ where $\alpha, D > 0$ are constants.
Take $\mathbb{E}$

$$d(\mathbb{E}X) = -\alpha(\mathbb{E}X)dt$$

Set $y(t) = \mathbb{E}(X(t))$

$$\begin{aligned}
&\Rightarrow \frac{dy}{dt} = -\alpha y \\
&\Rightarrow y(t) = x_0 e^{-\alpha t} = \mathbb{E}(X(t))
\end{aligned}$$

Ito's for $f(X) = X^2$:

$$\begin{aligned}
df &= (-\alpha X f' + D f'') + f'\sqrt{2D}dB \\
d(X^2) &= (-2\alpha X^2 + 2D)dt + 2X\sqrt{2D}dB
\end{aligned}$$

Take $\mathbb{E}$:

$$
\begin{aligned}
d(\mathbb{E}X^2) &= (-2\alpha\mathbb{E}X^2 + 2D)dt + \underbrace{\mathbb{E}(2X\sqrt{2D}dB)}_{\searrow 0 \text{ (Ito SDE)}} \\
\frac{d}{dt}\mathbb{E}(X^2) &= -2\alpha(\mathbb{E}X^2) + 2D \\
\frac{dy}{dt} &= -2\alpha y + 2D,\ y(t) = \mathbb{E}(X^2),\ y(0) = X^2
\end{aligned}
$$

3.7 Lecture 24 - Solving SDEs with Ito's Lemma

Recap:

- Ito's Lemma as stochastic chain rule
- Take ODE, $x(t) \in \mathbb{R}$

$$
\begin{aligned}
\frac{dx}{dt} &= u(x) \\
dx &= u(x)dt
\end{aligned}
$$

Then $f(x(t))$ satisfies

$$
\begin{aligned}
\frac{df}{dt} = f'(x)\frac{dx}{dt} &= f'(x)u(x) \\
\Rightarrow df &= f'(x)u(x)dt
\end{aligned}
$$

Take Ito-SDE, $X(t) \in \mathbb{R}$

$$dX = u(X)dt + \sqrt{2D}dB$$

Then $f(X(t))$ satisfies

$$df = (uf' + Df'')dt + f'\sqrt{2D}dB$$

Remarks: Types of ODEs we will encounter

- $\frac{dy}{dt} = \alpha y \Rightarrow y(t) = Ae^{\alpha t}$
- $\frac{dy}{dt} = \alpha y + \beta \Rightarrow y(t) = Ae^{\alpha t} - \frac{\beta}{\alpha}$
  where A is set by initial condition.

$\boxed{\text{ex}}$ BM with drift $X(t) \in \mathbb{R}$

$$dX = udt + \sqrt{2D}dB, X(0) = x_0$$
$$u, D \text{ are constants, } D > 0$$

Take the expectation $\mathbb{E}$:

$$\begin{aligned} d(\mathbb{E}(X)) &= udt \\ \Rightarrow \frac{d}{dt}(\mathbb{E}X) &= u \\ \Rightarrow \mathbb{E}(X(t)) &= ut + x_0 \end{aligned}$$

Ito:

$$df = (uf' + Df'')dt + f'\sqrt{2D}dB$$

Set $f(x) = x^2$

$$\begin{aligned} d(X^2) &= (2Xu + 2D)dt + 2X\sqrt{2D}db \\ d(\mathbb{E}X^2) &= (2u\mathbb{E}X + 2D)dt + \mathbb{E}(2X\sqrt{2D}dB) \\ \frac{d}{dt}(\mathbb{E}X^2) &= 2u^2t + 2ux_0 + 2D \\ (\mathbb{E}X^2) &= u^2t^2 + 2ux_0t + 2Dt + x_0^2 \\ \mathrm{Var}X &= \mathbb{E}X^2 - (\mathbb{E}X)^2 = 2Dt \\ &= \mathrm{Var}(\sqrt{2D}B(t)) \end{aligned}$$

Here we know the distribution of $X(t)$

$$\begin{aligned} X(t) &= \int_0^t dX = \int_0^t udt' + \int_0^t \sqrt{2D}dB + x_0 \\ &= ut + x_0 + \sqrt{2D}B(t) \\ \Rightarrow X(t) &\sim N(ut + x_0, 2Dt) \end{aligned}$$

$\boxed{\text{ex}}$ Geometric BM (GBM) for modeling stock prices, $X(t) \in \mathbb{R}$

$$dX = \mu X dt + \sigma X dB, \ X(0) = x_0$$
$$\text{with constants } \mu, \sigma$$

Find $\mathbb{E}X$

$$\begin{aligned}
d(\mathbb{E}X) &= \mu\mathbb{E}X dt \\
\frac{d}{dt}\mathbb{E}X &= \mu\mathbb{E}X \\
\mathbb{E}X &= x_0 e^{\mu t}
\end{aligned}$$

Find $\mathbb{E}X^2$ using Ito

$$df = (\mu X f + \frac{1}{2}\sigma^2 X^2 f'')dt + f'\sigma X dB$$

Set $f(X) = X^2$

$$\begin{aligned}
d(X^2) &= (2\mu X^2 + \sigma^2 X^2)dt + 2\sigma X^2 dB \\
d(\mathbb{E}X^2) &= (2\mu + \sigma^2)\mathbb{E}X^2 dt \\
\frac{d}{dt}\mathbb{E}(X^2) &= (2\mu + \sigma^2)\mathbb{E}(X^2) \\
\mathbb{E}X^2 &= x_0^2 e^{2\mu+\sigma^2}t
\end{aligned}$$

We can get the distribution of $X(t)$ from this.

Why GBM?
Ito with $f(X) = \log(\frac{X}{x_0}) = \log X - \log x_0$

$$\begin{aligned}
d(\log X) &= (\mu X \cdot \frac{1}{X} + \frac{1}{2}\sigma^2 X^2 \cdot (\frac{-1}{x^2}))dt + \frac{1}{X}\sigma X dB \\
&= (\mu - \frac{1}{2}\sigma^2)dt + \sigma dB \Rightarrow \text{BM with drift} \\
&\Rightarrow \log X \text{ is normal} \\
\mathbb{E}(\log X) &= (\mu - \frac{1}{2}\sigma^2)t + \log x_0 \\
\operatorname{Var}(\log X) &= \sigma^2 t \\
\log X(t) &\sim N\left((\mu - \frac{1}{2}\sigma^2)t + \log x_0,\ \sigma^2 t\right)
\end{aligned}$$

2d SDEs
$\boxed{\text{ex}}$ Particle $(X(t), Y(t)) \in \mathbb{R}^2$, moving in $2d$ with random kicks.

$$dX = u(X,Y)dt + \sqrt{2D_1(X,Y)}dB_1$$
$$dY = v(X,Y)dt + \sqrt{2D_2(X,Y)}dB_2$$

For functions $u, v, D_1, D_2 : \mathbb{R}^2 \to \mathbb{R}$ and $B_1(t), B_2(t)$ are independent BMs.
<u>2d Ito's Lemma</u>
Find df for $f(X,Y),\ f : \mathbb{R}^2 \to \mathbb{R}$ using Taylor's expansion.

$$\begin{aligned} df &= f(X+dX, Y+dY) - f(X,Y) \\ &= dX\partial_X f + dY\partial_Y f + \frac{1}{2}\begin{bmatrix} dX & dY \end{bmatrix}\begin{bmatrix} \partial^2_{XX} f & \partial^2_{XY} f \\ \partial^2_{YX} f & \partial^2_{YY} f \end{bmatrix}\begin{bmatrix} dX \\ dY \end{bmatrix} + \mathcal{O}(dt^{\frac{3}{2}}) \end{aligned}$$

Ito's Lemma:

$$df = \left(u\frac{\partial f}{\partial X} + v\frac{\partial f}{\partial Y} + D_1\frac{\partial^2 f}{\partial X^2} + D_2\frac{\partial^2 f}{\partial Y^2} \right) + \frac{\partial f}{\partial X}\sqrt{2D_1}dB_1 + \frac{\partial f}{\partial Y}\sqrt{2D_2}dB_2$$

"Add the Ito's lemma expressions for each SDE"

$\boxed{\text{ex}}$ Microbe walking on a line $X(t) \in \mathbb{R}$

$$dX = v\cos\theta dt + 0,\ X(0) = 0$$
$$d\theta = 0 + \sqrt{2D}dB,\ \theta(0) = 0$$

- Particle tries to move in θ direction.
- 2d active particle $\to$ 3 SDEs
- v, D are constants, $D > 0$

What is $\mathbb{E}X$?

$$d(\mathbb{E}X) = v\mathbb{E}(\cos\theta)dt$$

Ito's Lemma for $f(X, \theta) = \cos\theta$

$$\begin{aligned}
df &= \left(v\cos\theta \frac{\partial f}{\partial X} + 0\frac{\partial f}{\partial \theta} + 0\frac{\partial^2 f}{\partial X^2} + D\frac{\partial^2 f}{\partial \theta^2} \right) dt + \frac{\partial f}{\partial \theta}\sqrt{2D}dB \\
d(\cos\theta) &= -D\cos\theta dt + \sin\theta\sqrt{2D}dB \\
\frac{d}{dt}(\mathbb{E}\cos\theta) &= -D\mathbb{E}(\cos\theta) \\
\mathbb{E}\cos\theta &= e^{-Dt} \quad (\theta(0) = 1) \\
\frac{d}{dt}\mathbb{E}X &= ve^{-Dt} \\
\mathbb{E}X(t) &= \frac{-v}{D}e^{-Dt} + \frac{v}{D} \quad X(0) = 0
\end{aligned}$$